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Blockchain-Based Governance: Implications for Organizational Boundaries and Structures

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Blockchain has emerged as a key Industry 4.0 technology, enabling novel forms of governance and coordination mechanism among organizations and markets. However, extant literature has largely focused on the technical aspects of blockchain, with limited attention to the behavioural and institutional aspects. In this paper, we argue that blockchain-based smart contracts and decentralized autonomous organizations represent the potential for a radical departure from traditional forms of contractual governance and hierarchy, carrying profound implications for the design and governance of economic transaction and organizational structures. We elucidate how blockchain technologies, characterized by transparency, immutability, programmability and decentralization, reduce transaction costs and establish an industrialized and trustless transactional governance system. Finally, we present an agenda for future research, highlighting the need for new theoretical frameworks and empirical evidence to understand the impact of blockchain on organizational design and forms.

Introduction

What determines the boundary of a firm remains a fundamental question for organizational scholars dating back to seminal works by the likes of Coase (1937) and Williamson (1975). Firms were conceptualized as an alternative mechanism for resource allocation, distinct from the allocation of resources facilitated by prices and contracts. However, the advent of cutting-edge technologies, including blockchain, artificial intelligence (AI) and machine learning (ML), represents more than just a leap in the efficiency of human problem-solving; it signals a potential upheaval in the very foundations of traditional business governance and organization. These technologies have ushered in a paradigm shift, offering capabilities that extend far beyond mere optimization – they have the capacity to fundamentally reshape the landscape of how business transactions are organized and conducted. In this context, blockchain emerges as a groundbreaking force, holding the promise to revolutionize not only how organizations operate internally but also how they interact within the broader market ecosystem.

Four key features of blockchain make it unique in transforming the organization of transactions between firms and markets – transparency, immutability, programmability and decentralization (Narayanan and Clark, 2017). We will describe these key features in detail in the next section. However, it is important to note that the transparency of blockchain systems, coupled with their decentralized nature, makes them resistant to censorship and tampering, and well suited for a variety of applications (Narayanan and Clark, 2017). These distinctive attributes, coupled with the wide array of applications of blockchain in

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the organizing of economic transactions, raise critical questions about the economic rationale of organizational boundaries and the design of internal organizational structures and processes (Lumineau, Wang and Schilke, 2021; Murray *et al.*, 2021; Vergne, 2020; Wang, Lumineau and Schilke, 2022).

Despite its many benefits, organizations face significant barriers to the successful implementation of blockchain solutions, with many early implementations being either failed or disappointing (Sodhi *et al.*, 2022). Without a well-established theoretical foundation to guide our understanding of blockchain's impact on organizations, various disconnected predictions emerge – some even heralding the demise of traditional organizations, while positioning technology as a panacea for all organizational inefficiencies. It becomes increasingly apparent that our current organizational theories possess limited explanatory capacity when it comes to understanding the profound impact of emerging technologies on the essence and structure of organizations (Puranam, Alexy and Reitzig, 2014). As Puranam, Alexy and Reitzig (2014) note, 'our existing organizational theories are deeply rooted in a context that no longer aligns with the realities of the contemporary world'. This emphasizes the pressing need for the creation of new, evidence-based theories that can effectively elucidate the various ways in which diverse blockchain solutions will influence organizational operations. These theories should also guide organizations in understanding which specific blockchain solutions are most appropriate for addressing their unique challenges and provide critical insights into the decision-making process for selecting and implementing blockchain solutions.

In this commentary, we delve into how blockchain is expected to fundamentally impact the organization of business transactions via markets and the relevance of organizational hierarchy. We underline the key features of blockchain technology that set it apart from other existing technologies, and how this is expected to influence how business transactions are organized. Furthermore, we shed light on how blockchain is expected to bring about transformative changes in the structure of organizations, particularly in terms of decision-making hierarchies, making them more structured and decentralized. We conclude with some directions for future research in this rapidly evolving field.

Blockchain fundamentals

Blockchains are defined as 'cryptography-based decentralized and distributed systems consisting of an ongoing list of digital records that are shared within a peer-to-peer network' (Wang, Lumineau and Schilke, 2022). Blockchains were introduced in 2008, with Bitcoin as the first decentralized peer-to-peer version of digital currency. However, the underlying concept of a 'cryptographically secured chain of blocks for storing information' existed before, in academic literature of the 1990s, particularly in the works of Haber and Stornetta (1991). Haber and Stornetta's (1991) work on document timestamping and data structures can be traced in blockchain's immutable property, as highlighted by Nakamoto in his whitepaper (Narayanan and Clark, 2017).

The development and maturation of blockchain technology can be broadly classified into three phases. Generation 1.0 blockchains aimed to address the inefficiencies of traditional monetary systems through bitcoins. The key structural distinction that differentiates blockchain from the legacy system is its decentralized nature, which facilitates transactions between parties without going through a central system. In this decentralized and distributed ledger system, no single node has exclusive rights to the ledger, but everyone has access to it through the Internet (Lumineau, Wang and Schilke, 2021). Generation 2.0 blockchains added a software layer over the blockchain, which could be used to code 'conditional logic' on blockchains. Smart contracts represent this advancement, wherein business logic can be encoded onto the blockchain system and then can self-execute transactions once the coded conditions are met. However, these blockchain technologies suffered from limited scalability features and were constrained in their ability to interact with each other. Generation 3.0 blockchains focused on addressing scalability and interoperability issues to enable mass adoption and development of blockchain as a platform. Key projects that added scalability and interoperability features on blockchains include Ethereum 2.0, Cardano and Polkadot. These projects were based on distributed ledger technology (DLT), which leverages data

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structures based on innovative validation and voting mechanisms to improve transaction processing latency.

Blockchains have four distinctive features – decentralization, transparency, immutability and programmability – that will have transformative effects on organizations. First, blockchain operates as a decentralized ledger, free from the control of central institutions, where each participant within the network maintains an identical copy of the ledger. Second, the intrinsic characteristic of decentralization fosters a transparent and widely shared data source, ushering in an unprecedented level of openness and accessibility (Davidson, De Filippi and Potts, 2018). Third, blockchain employs a unique mechanism where transaction data are encrypted and organized into blocks. These blocks are then securely linked together, forming an unalterable and chronological transaction trail, hence the nomenclature 'blockchain' (Narayanan and Clark, 2017). This immutability is at the core of blockchain's integrity and reliability. Fourth, blockchain technology incorporates 'smart contracts', which are self-executing agreements between trading partners and come with programmable capabilities. Smart contracts empower registered peers within the network to validate transactions autonomously and anonymously (Davidson, De Filippi and Potts, 2018). Moreover, they play a pivotal role in determining whether a new block can be appended to the blockchain in a precise chronological order, enhancing the system's efficiency and trustworthiness (Lumineau, Wang and Schilke, 2021).

Initially adopted as a technology tool for improving business processes, blockchains have evolved beyond being merely efficiency and productivity-enhancing technology. Current blockchain solutions, like smart contracts and decentralized autonomous organizations (DAOs), are increasingly positioning blockchains as a platform for interfirm cooperation and collaboration. We discuss these briefly here.

Smart contracts

Smart contracts are programs stored on a blockchain that execute when predefined conditions are met (Murray *et al.*, 2021). Unlike bitcoins, smart contracts have built-in programming interfaces and an additional software layer on top of the blockchain. Ethereum-based scripting languages such as Solidity allow for the automation of traditional contracts to facilitate, verify or enforce contract terms. They also demonstrate that certain conditions were met before executing the outcome/payment. Thus, smart contracts automate agreement execution and dictate business rules between parties without intermediary supervision. Fundamental features of blockchain – like decentralized consensus, machine-based automation and immutability – ensure that smart contracts remain unchanged and indelible. These features provide the necessary trust in smart contracts, because any data manipulation will lead to the breakdown of the blockchain system (Davidson, De Filippi and Potts, 2018; Murray *et al.*, 2021).

Decentralized autonomous organizations

DAOs emerged in 2016 with the aim of creating organizations devoid of managers and hierarchies, replacing them with automated smart contracts (Ellinger *et al.*, 2023). DAOs are 'pseudo-legal' organizations, operated by humans and programs. Essential organizational features – like governance and functioning – are coded on the Ethereum blockchain, which is decentralized and responds to inputs according to the programming logic built on them. DAOs automatically respond to inputs as per the pre-programmed logic, and initiate actions that are immutable. The set of output actions ranges from distributing financial rewards to undertaking computations and controlling the actions of physical systems like electromechanical devices. DAOs are funded through digital blockchain-based tokens and control in the organization is managed according to the stakes contributed; that is, entities with higher contributions receive higher decision weights in organizational decision-making. In addition, any stakeholder in the DAO can bring proposals to be voted on by other token holders, thus virtually controlling decision-making.

Blockchain and organizations

In this section, we analyse the potential impact of blockchain in transforming firm boundaries, the organization of transactions and organizational structures and forms.

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Transaction costs and organizing via markets or hierarchy

The fundamental inquiry into the existence, scope and boundaries of a firm is rooted in the concept of transaction costs that firms encounter when utilizing the free market to organize their business activities (Coase, 1937; Williamson, 1975). Building upon Coase's (1937) insights into transaction costs, Williamson (1975) highlighted that organizational actors aim to optimize the benefits of interdependence by selectively assigning transactions, which vary in their characteristics, to different governance structures (Williamson, 1985, p. 18). Bringing in the behavioural assumptions of bounded rationality and opportunism in the formal analysis, Williamson (1975) argued that transactions are assigned to governance structures based on three factors: uncertainty, asset specificity and the frequency of transactions. Thus, transaction costs determine when certain transactions are better conducted within hierarchical structures rather than through market mechanisms.

Organizational theory research has recognized the role of technology in fundamentally shaping the boundaries of firms (Chandler, 1962; Coase, 1937, p. 397; Menz *et al.*, 2021). For example, Coase (1937) himself highlighted the significance of the telephone in shaping firm boundaries. Similarly, Chandler (1962) discussed how electrical and chemical technologies from the late nineteenth century led to the consolidation of industries into large, vertically integrated companies and the development of the multidivisional form (M-form) structure. Additionally, the emergence of the factory, as noted by McCraw (1998), was closely tied to the Industrial Revolution and the ability of firms to harness water and steam power in a single location.

The rise of information and communication technologies (ICTs), especially the Internet, had a profound effect on improving the efficiency of markets by reducing transaction costs, which in turn led to a greater reliance on markets, as opposed to hierarchies, for the coordination of economic activities (Gurubaxani and Whang, 1991). In this way, improvements in ICTs paved the way for the decentralization of many organizational functions. Substantial research has been conducted building on these propositions, with the thesis that technological improvement will result in an overall shift towards the use of markets rather than hierarchies to coordinate economic activity.

Similar to existing ICTs, blockchain also offers the functionality to further shift the organizational boundary towards the market by reducing transaction costs (Catalini and Gans, 2018). For instance, blockchain can enhance collaboration between a firm and its partner network through immutable record-keeping and data verification by multiple nodes, fostering trust in information exchange within the network. Information shared over the blockchain relies on consensus algorithms and undergoes independent verification by each node, facilitating information exchange between parties. Additionally, with each transaction timestamped, any attempt by a network member to alter information can easily be tracked with evidential support (Wang, Lumineau and Schilke, 2022). Blockchains are also extensively used in monitoring applications in market transactions. Blockchain's decentralization and immutability help address a major challenge firms face when transacting through markets: the limitation in attaining traceability and transparency in their supply chain due to the complex movement of products across multiple geographical regions before reaching consumers (Sodhi *et al.*, 2022).

However, when arguing for blockchain as a potential solution to reduce transaction costs in many settings, an important question arises: why do we need blockchain if we have existing digital technologies offering the same functionality of improving market-based organizing through the minimization of transaction costs? Here, it is important to underline that previous and other current technologies have primarily focused on improving information provisioning and data exchange (Lumineau, Wang and Schilke, 2021). Existing technologies like the Internet, the Internet of Things, AI and ML have facilitated the sharing of digital documents, enhanced predictability through intelligent analytics, enhancing efficiency and lowering costs in information exchange in large organizations (Ciulli and Kolk, 2023; George and Schillebeeckx, 2022; Makarius *et al.*, 2020). However, these technologies often leave a critical issue unresolved – opportunism and trust issues-based drivers of transaction costs (Gaur *et al.*, 2022; Singh and Gaur, 2021). Thus, despite the efficiency brought in by existing technologies, the transaction costs arising from partner opportunism and

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mistrust limit organizations' ability to utilize the efficiencies offered by the market.

For example, consider the use of radio-frequency identification (RFID) technology in supply chain management. RFID tags are used to track and manage inventory in a highly efficient manner. These tags can store information about products, such as their origin, manufacturing date and shipping history, and this data can easily be shared between different parties in the supply chain. While RFID technology streamlines inventory management and improves transparency, it still relies on traditional contracts and legal recourse in case of disputes. If, for example, a shipment of perishable goods arrives late and spoils, the parties involved would need to resort to legal procedures to resolve the issue. The enforceability of digital contracts and transactions still relies on traditional legal systems, introducing a level of uncertainty in the event of disputes or non-compliance.

It is this gap between the market and the hierarchy that blockchain solutions like smart contracts potentially address. Smart contracts leverage codes and algorithms to enforce agreements, thus building trust, enhancing communications, reducing disputes and preventing opportunism (Ellinger *et al.*, 2023). Smart contracts offer a unique way of governing collaborations through a self-contained system of formal rules, defined by a set of protocols. Blockchains promote cooperation by preventing opportunistic behaviour from the outset (Wang, Lumineau and Schilke, 2022). Any actions that violate the pre-agreed rules are simply ignored and not executed. For instance, fraudulent attempts to tamper with blockchain records are swiftly rejected because they do not conform to the consensus protocol. Similarly, failure to fulfil contract obligations after a partner completes their tasks is eradicated, thanks to smart contracts that automatically execute agreements upon successful verification (Davidson, De Filippi and Potts, 2018; Wang, Lumineau and Schilke, 2022).

Blockchain provides a system based on impersonalized trust for organizing economic activities via the market. Thus, with the emergence of blockchain technology, we are likely to witness a further shift towards market-based organizing through reduced transaction costs (Catalini and Gans, 2018). Blockchain's ability to enhance collaboration, ensure trust through consensus algorithms and provide traceability (Wang, Lumineau and Schilke, 2022) demonstrates its capacity to address previously unresolved challenges associated with opportunism and trust in digital technologies (Lumineau, Wang and Schilke, 2021; Wang, Lumineau and Schilke, 2022).

Agency cost and decentralized and distributed organizations

The adoption of hierarchical organizational structures serves as a strategic response to the multifaceted challenges posed by transaction costs, emanating from variables encompassing the frequency of interchanges, asset specificity and inherent uncertainty (Cyert and March, 1963). It is worth noting that even with well-designed incentive systems, aligning the behaviour of agents within hierarchical structures can be challenging due to the existence of multiple possible equilibriums in group interactions (Dalton *et al.*, 2007). Extensive research in the field of organizational theory has shed light on the intricate relationship between organizational outcomes and internal structures, such as the delegation of authority, information provisioning, knowledge and incentives among individuals engaged in the various facets of the organizational production process (Collis and Montgomery, 1997). Amongst these, information provisioning and delegation of authority are two variables which have been profoundly impacted by advances in ICTs and associated changes in organizational structure and design (Menz *et al.*, 2021).

To grasp the potential impact of blockchain on organizational structure and design, it is crucial to focus on two core elements of organizational design: the way information is exchanged within the organization and how decision-making authority is distributed (Vergne, 2020). The information exchange network within an organization can vary widely. At one end of the spectrum there is a highly centralized model, where each unit operates independently, acquiring and processing information without external interaction. At the other end, there is a distributed information exchange model, characterized by widespread communication across the organization. Likewise, decision-making can take two primary forms: concentrated or distributed. In a concentrated model, decision-making authority primarily resides at the top of the hierarchy, with lower levels mainly involved in information processing. In contrast,

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the distributed model involves delegating decision-making to various hierarchical levels (Vergne, 2020).

There are technologies such as AI and ML that offer opportunities for more decentralized knowledge-sharing among members and units. However, they do not inherently guarantee the distribution of decision-making authority (Vergne, 2020). Blockchain solutions, such as smart contracts, offer a path forward for achieving both decentralization and distribution of decision-making. Blockchain's decentralized nature ensures data availability to every organizational node, promoting decentralization. Simultaneously, smart contracts introduce programmability, creating a trustless system and enabling the automation of decision-making processes (Wang, Lumineau and Schilke, 2022). Unlike a centralized hierarchy, decisions can be delegated to unit levels with full visibility for all organization units. This presents a solution to minimize potential principal–agent problems, as decisions align with pre-approved conditions. DAOs represent an extreme form of organization through blockchains. They offer the opportunity to codify organizational structure and automate decision-making that was traditionally handled by managers (Murray *et al.*, 2021). The role of managers within the organization would be to handle dynamic and complex tasks, instead of routine decision-making. Thus, blockchain offers the possibility of creating a more decentralized and distributed organization.

Future research directions

Given the limited empirically driven insights on blockchains, more research is required to identify the organizational problems that can be addressed by blockchain using practical, real-life cases. This will help in identifying the challenges faced by organizations in the successful implementation of blockchain solutions. Identifying the factors that lead to both successful and failed implementations is critical for practical adoption.

Current applications of blockchain essentially view it as a supply chain optimization tool. However, our discussion highlights that its applications go far beyond traditional efficiency-enhancing technologies. To develop a more nuanced understanding of how organizations can benefit from blockchain, it would be important to focus on organizational design and governance issues and address the scalability challenges associated with blockchain adoption. A joint effort by organizational scholars and technology experts needs to be directed towards understanding how organizations handle increased transaction volumes as they adopt blockchain solutions and evaluate the scalability limits of existing blockchain platforms. From a corporate strategy perspective, we need to understand how blockchain adoption contributes to the competitive advantage of organizations. This includes analysing whether early adopters of blockchain technology gain a competitive edge over their peers. We elaborate on these threads for future research below.

Decision between governance modes – market versus hierarchy

Generation 1.0 blockchain solutions typically offer to reduce transaction costs in information sharing, cut administrative costs, optimize lead time and minimize risk through decentralized monitoring of the entire transaction value chain. A prime illustration of this is the Food Trust platform by IBM, which harnesses blockchain technology to enhance the traceability of the food supply chain. IBM Food Trust offers instant access to actionable data and insights to farmers, food producers and all participants in the food supply chain. Global corporations such as Walmart, Carrefour and Nestlé have integrated IBM Food Trust into their food supply chains. A smart contract layer over the Food Trust platform allowed for the creation of automated processes for the approval of payments and updating inventory in real time. Beyond optimizing transaction costs, blockchain technologies like smart contracts and DAOs offer the possibility to significantly alter organizational boundaries by addressing issues of opportunism and appropriability. Opportunism arises from actors not sharing full information or providing objective assessments of likely outcomes or behaving cooperatively during the execution of economic exchange (Williamson, 1985). Blockchain, by limiting the need for trust in transactions, offers the potential to address issues of opportunism in economic exchange (Wang, Lumineau and Schilke, 2022). However, we do not have enough understanding of how different blockchain solutions can help solve specific organizational governance problems. Research in this

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area should also aim to identify boundary conditions that limit the usability of current blockchain features. For example, current blockchain solutions are not capable of helping to resolve the opportunism problem at contract renewal and other behavioural uncertainties that are more difficult to design into contracts or automate in smart contracts.

Design of organizational hierarchy

The design of organizational hierarchy hinges on two fundamental challenges: the division of labour and the integration of efforts to achieve a shared objective (Stinchcombe, 1965). While the structure of an organizational hierarchy is typically based on the assumption that the coordinated efforts of its members will propel the organization towards its goals, Salznick's (1957) classic analysis suggests that the organization's goals can evolve independently of its explicit objectives. Furthermore, there is an underlying assumption that, even with the presence of informal organizations within the formal hierarchy, the contributions of its agents will align with the organizational goals (Puranam, Alexy and Reitzig, 2014).

This perspective of organizations as conflicted coalitions highlights the challenging dilemma in designing organizational hierarchies: the strategic arrangement of control and trust mechanisms to mitigate agents' opportunistic behaviour, foster unity and predictability, and achieve optimal performance. With the emerging potential for the digitization of trust and the automation of decision-making based on consensus, fundamental questions arise on how to incorporate these into organizational designs (Ellinger *et al.*, 2023; Lumineau, Wang and Schilke, 2021).

The rise of DAO-based autonomous and decentralized organizations, such as MakerDAO and Aragon, presents the possibility of using blockchain technology to address trust and control issues in organizational design (Ellinger *et al.*, 2023). DAOs offer an alternative approach to organizing, built on collective ownership and autonomous contributors through tokenization (Murray *et al.*, 2021). Additionally, DAOs rely on the voluntary contributions of their community members to sustain ongoing operations (Ellinger *et al.*, 2023). What sets this model apart from current organizational designs, which are based on centralized ownership and governance, is its governance without a central authority (Wang, Lumineau and Schilke, 2022). In this context, DAOs represent an innovative 'organizational form'. They constitute a multi-actor system with clear boundaries and a collective system-level purpose, to which each participant is expected to contribute their efforts (Ellinger *et al.*, 2023; Lumineau, Wang and Schilke, 2021).

Future research may explore the potential of integrating DAOs into existing organizational structures, creating a hybrid of hierarchy and DAO autonomy. While DAOs are currently in a proof-of-concept stage, further research is needed to address the challenges of scaling these solutions for large-scale use in organizational decision-making. Ensuring the reliability and dependability of DAOs in upholding agreements and their role as integral components of an organization's structure and coordination mechanism depends on a combination of internal organizational factors and external variables (Ellinger *et al.*, 2023). These external factors encompass the accessibility of liquidity and the secure operation of the underlying blockchain technology to reduce the vulnerability to security breaches and other forms of attack.

Implications for boundaries and the design of multinational enterprises

The international business field has extensively explored the existence of multinational enterprises (MNEs) and their foreign market entry strategies. Internationalization theory has become the widely accepted framework for understanding MNEs. This theory builds on Coase's (1937) work, emphasizing the comparative costs of operating through markets across national borders, with hierarchical structures favoured for minimizing transaction costs in cross-border operations. Buckley and Casson (1976) further underscore that production entails a network of interconnected activities and the exchange of intermediate products, including materials and intangible assets like knowledge. While efficient coordination relies on well-functioning markets, many encounter imperfections such as pricing difficulties and government interventions. This is especially critical in knowledge-intensive domains like research and development (R&D). To address these challenges, activities often consolidate under common ownership, leading to the formation of MNEs when operations span

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different countries. Transaction costs thus underpin the hierarchical structure of MNE subsidiaries.

Furthermore, the internalization perspective views MNEs as approximations of perfect markets, with their internal structures designed to convey shadow prices to managers across decentralized subsidiaries, all with the aim of maximizing overall MNE profits (Buckley and Cason, 1976). Building on our previous discussion regarding the impact of blockchain adoption on minimizing transaction costs, future research can explore the implications of blockchain for MNEs' entry mode choices, such as greenfield investments, licensing or joint ventures. Additionally, emerging research highlights internal market-like structures within business organizations operating on principles of price mechanisms and property rights. Future research could investigate how blockchain facilitates the replication of such internal price mechanism-driven structures within MNEs.

Blockchain, with its emphasis on 'digitalizing and industrializing trust', presents another crucial avenue for future research in international business. Examining how the removal of the need for trust in transactional relationships through blockchain affects control structures within MNE–subsidiary and MNE–supplier relationships (Casey and Vigna, 2018) holds promise for advancing our understanding of this dynamic field. For instance, a multinational enterprise may leverage blockchain to streamline its supply chain operations, allowing for real-time tracking of goods and payments, reducing the need for intermediaries and enhancing trust between the parent company and its global network of suppliers and subsidiaries. If blockchain adoption can minimize the trust deficit that MNEs face in distant markets, what are the implications for research that examines the impact of different distance measures (e.g. cultural, institutional, geographic, etc.) on MNE strategic choices (e.g. entry mode, location, timing, etc.) and performance? Additionally, in the face of rising anti-globalization sentiments and the increasing adoption of decoupling as a policy in many countries, the role of blockchain in minimizing trust issues becomes particularly crucial (Cason, 2021; Meyer *et al.*, 2023; Witt *et al.*, 2023). MNEs can harness the power of blockchain to manage their operations across various nations in a secure, transparent and efficient manner without the need to have distributed physical operations. However, it is essential to note that the use of blockchain as a solution for cross-border business organizing by MNEs would require a global consensus on technical standards and regulatory policy.

Confluence of blockchain and other Industry 4.0 technologies

At its core, information processing has been regarded as the key factor influencing organizational hierarchy decisions (Cyert and March, 1963; Simon, 1997; Thompson, 1967). As posited by Simon (1997, p. 240), 'in a post-industrial society, the key problem in research related to organizational structure is how to organize to make decisions—that is, to process information'. Similarly, bounded rationality and cognitive limitations are regarded as the key factors contributing to the specific design of organizational hierarchy (Cyert and March, 1963). Prior literature on organizational design has fundamentally examined whether the difference in the way organizations are structured can be explained by the difference in the manner and nature of information processing (Joseph and Gaba, 2020). Furthermore, research in this direction shall take a broader perspective of the technology portfolio and analyse the impact of profit-driven technologies such as AI/ML and blockchain and their implications for organizational design, especially given their complementary aspects of transformation, with AI/ML paving the way for decentralized organizations and blockchain supporting distributed organizational boundaries.

Conclusion

In conclusion, the integration of blockchain technology into the realm of organizational governance has ushered in a new era of possibilities and challenges for organizational research. As we elucidated, blockchain's potential goes beyond mere efficiency enhancement, extending to the transformation of traditional structures and trust mechanisms. Future research should delve into the intricate dynamics of blockchain's impact on decision-making processes, exploring the scalability challenges, competitive advantages and potential coexistence of decentralized

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autonomous organizations with established hierarchies. Further, the confluence of blockchain with other Industry 4.0 technologies, such as AI/ML, marks a pivotal moment in the evolution of organizational design, promising both disruption and innovation. As organizations adapt to this rapidly changing landscape, it is clear that understanding and harnessing the full potential of Industry 4.0 technologies will be a critical endeavour for scholars and practitioners alike.

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